

Hibbeler Crib Sheet vs. R Flori

Semicircular area

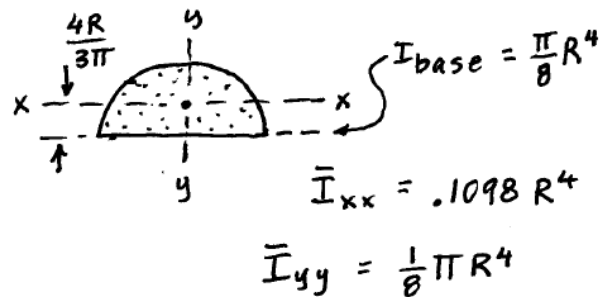
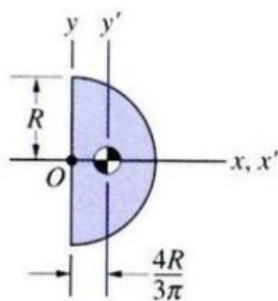
$$\text{Area} = \frac{1}{2} \pi R^2$$

$$I_{xy} = 0$$

$$I_{x'y'} = 0$$

$$I_x = I_y = \frac{1}{8} \pi R^4$$

$$I_{x'} = \frac{1}{8} \pi R^4, \quad I_{y'} = \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) R^4,$$



Quarter-circular area

$$\text{Area} = \frac{1}{4} \pi R^2$$

$$I_x = I_y = \frac{1}{16} \pi R^4$$

$$I_{xy} = \frac{1}{8} R^4$$

